

VIKAS DAS

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4 years of experience designing, scaling, and optimizing microservices using Java, Spring Boot, and Apache Kafka, serving 700+ enterprise clients at 500K+ user scale. Strengths in event-driven architecture, distributed concurrency, enterprise migrations, and database tuning across PostgreSQL, Redis, and Elasticsearch. Increasingly focused on AI agent tooling, building custom MCP servers over production datastores to automate reporting and production diagnostics.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Shell Scripting (Bash)

Backend & Frameworks: Spring Boot, Spring Security, Hibernate, JPA, Microservices Architecture, RESTful APIs, System Design, JUnit, Mockito

Databases & Search: PostgreSQL, Redis, Elasticsearch, MongoDB, MySQL, MariaDB

Messaging & Concurrency: Apache Kafka, Event-Driven Architecture, Distributed Locking, Virtual Threads, Asynchronous Processing

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, Nginx

Observability & Tools: Prometheus, Grafana, ELK Stack, New Relic, Git, GitLab, Postman, SonarQube, HashiCorp Vault, Consul

AI & Developer Productivity: Model Context Protocol (MCP), Custom MCP Servers, LLM Tool Integration, Context Engineering, Prompt & Token Optimization, AI-augmented development workflows, automated code migration tooling

EXPERIENCE

Peoplehum Pvt Ltd.

Senior Software Engineer

Bengaluru, India

Nov 2025 – Present

- Owned end-to-end development of the Attendance, Leave, and Performance modules serving 700+ enterprise clients (500K+ users), driving architectural decisions while mentoring 5 engineers through structured code reviews and design pattern workshops.
- Migrated Elasticsearch cluster from v6 to v8 and implemented cursor-based pagination (`search_after`), enabling queries across 100M+ records in under 100ms while eliminating deep-pagination memory bottlenecks.
- Redesigned attendance calculation from scheduled batch jobs to an event-driven architecture, computing 5-6M daily records with sub-second freshness in place of hourly batch delays.
- Built custom MCP (Model Context Protocol) servers exposing read-only production MariaDB and MongoDB to AI agents, enabling natural-language report and data extraction and eliminating manual query writing for support and analytics requests.
- Developed an MCP-based production log inspection tool routed to a lighter LLM, and authored versioned architecture and flow context files that removed repeated full-codebase scans, cutting token consumption and turnaround time per investigation.
- Implemented distributed locking using Redis and Kafka event synchronization for cross-subsidiary employee transfers, maintaining transactional consistency across payroll, leave, and attendance microservices.
- Led the Java 21 upgrade initiative, personally migrating 30+ microservices while establishing migration patterns and tooling adopted across 100+ services, improving throughput on latency-bound services by up to 35%.
- Architected a multi-tenant Break Management module standardizing policy enforcement across 15+ shift configurations, and shipped a self-service attendance regularization workflow now processing 5,000+ requests monthly without administrator involvement.

Thermo Fisher Scientific

Software Engineer

Bengaluru, India

Dec 2024 – Nov 2025

- Engineered a high-throughput Kafka synchronization microservice to maintain data consistency across distributed databases, decreasing overall API latency by 30%.
- Optimized complex PostgreSQL queries and integrated strategic Redis caching layers, cutting database CPU utilization by 50%.
- Provisioned stateful Redis clusters on Kubernetes, increasing resource efficiency and reducing monthly QA cloud expenditures by 40%.
- Configured real-time observability pipelines with Prometheus, Grafana, and ELK Stack, reducing Mean Time to Recovery (MTTR) by 45%.

FlexEHR Healthcare Solutions

Software Development Engineer I

Bengaluru, India

Dec 2022 – Dec 2024

- Built and maintained high-concurrency RESTful microservices processing 10,000+ daily healthcare transactions while sustaining 99.9% uptime.
- Refactored database indexing and execution plans in PostgreSQL, accelerating query retrieval performance by 60%.
- Strengthened HIPAA and SOC2 compliance across 9 microservices by implementing HashiCorp Vault for secrets management and Consul for dynamic service discovery.
- Constructed a bulk CSV data ingestion pipeline, eliminating 15 hours of manual data entry work for operational teams weekly.
- Integrated WhatsApp Cloud API for programmatic appointment notifications, boosting patient engagement by 40% and decreasing missed appointments.

Software Engineering Intern

May 2022 – Nov 2022

- Developed 3 core EHR microservices (Pharmacy Management, Garuda notifications, patient intake) using Creational Design Patterns (Factory, Builder) to modularize business logic across 15+ API endpoints.
- Fine-tuned MongoDB collection indexing strategies, reducing average document retrieval response times by 25%.

PROJECTS

Webisssh | Java, Spring Boot, OpenAPI

[GitHub](#)

- Built a self-hosted Swagger alternative that scans Spring Boot controllers to auto-generate OpenAPI documentation, with folder-based organization for managing multiple environments and services from one browser console.

PropT Conrad | React, Node.js, PostgreSQL

[Live](#)

- Built an end-to-end property management platform featuring rental listings, second-hand marketplace, and real-time notification system for tenant-landlord communication.

EDUCATION

Savitribai Phule Pune University

Pune, India

Bachelor of Engineering in Information Technology (CGPA: 8.35/10)

Aug 2019 – May 2023

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Object-Oriented Programming (OOP), Software Engineering, Distributed Systems